AMNS/BAH/720-SCB/0322-03/263315

DESIGN, ENGINEERING, SUPPLY, ERECTION, INSTALLATION, AUTOMATION & COMMISSIONING OF SAFETY FANCING FOR VARIOUS AM/NS LOCATIONS.

ArcelorMittal Nippon Steel India

SAFETY STEEL MESH FENCING OF PROCESS LINE	AM/NS INDIA	A joint venture between ArcelorMittal and Nippon Steel Corporation ArcelorMittal NIPPON STEEL	REFRENCE NO-

SCOPE OF WORK FOR DESIGN, ENGINEERING, SUPPLY, ERECTION, INSTALLATION, AUTOMATION & COMMISSIONING OF STEEL MESH SAFETY FENCING & PIPE BARRICATION GAURDING OF PROCESS LINES, TRANSFER TROLLY & SOLAR PANEL AREA STEEL MESH SAFETY FENCING FOR ALL SSC

AS PER SOW ARCELORMITTAL NIPPON STEEL INDIA LTD BAHADURGARH.

The following is the BOQ. In respect of terms & conditions, not explicitly provided in this Special Terms & Conditions, provisions of General Conditions of contract are applicable.

Bidders are required to go through the tender document carefully before giving final quote for items mentioned in the attached list. If required, bidders can also visit our site for understanding the job properly; however, they should inform their program for site visit in advance to the contact persons (preferable technical contact 1).

1. Complete Modular, Extra Heavy-Duty Machine Safety Fence for Machine Guarding.
2. The Perimeter Protection system consisting, uprights 60X60X2 mm and panel with frame 40x40x1.5 mm & fence grid of 22x100x3 mm.
3. Complied of safety standard ISO-13857: Safety distances to prevent the contact of dangerous Areas with upper and lower limbs
4. Complied of safety standard ISO 13854- EN 349: Minimum gaps to avoid crushing parts of the human body.
5. Impact test proven machine guarding fence as per ISO 14120.
6. Material: IS 2062 E250 & SAE 1008 Steel used to ensure the structure strength.
7. Pre-galvanized and powder coated safety fence.
8. Square Pole shade is standard of RAL: Yellow 1021, Steel Mesh Shade is standard of RAL: black 9005.
9. Zinc Plated Hardware with rubber capping of uprights.
10. Wedge type anchor fastener with Chemical Anchoring for the Uprights Fixing.
11. Anti-Climbed design of Wire Mesh.
12. CAD design to be submit for approval before manufacturing of product.
13. No cutting and Welding (Hot Work) Required.
14. Providing all labour for erection, installation, termination, commissioning for line wise safety fencing.
15. Providing additional accessories for Steel mesh safety fencing to complete project successfully.
16. Providing all required cables, wires, electric fitting, & fitting arrangement.

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17. Vendor shall include service & commissioning team to & fro charges, conveyance charges, lodging boarding charges.
18. All work to be carried out as per safety norms which shall be guided to AM/NS team during work & no tolerance in safety will be accepted and penalty to be imposed if any violation.
19. Arranging the proper housekeeping after daily work completion & not leave the work site in an unsafe condition or any other condition that might cause injury to personnel, damage to existing work, plants or equipment.
20. Verify all installed equipment's before declared the work completion and should be certified by user.
21. Providing the individual equipment bill warranty period from date of installation.
22. Free servicing and repairing of installed Equipment's during warranty periods.

Location	Job Description	Qty	Remark
Bahadurgarh SC	CRNCTL Line, CRS Line, CRWCTL Line, HRNCTL Line, HRS Line, Transfer trolley, Solar Inverter room fencing	8 Set	• 1.5 Meter Height Steel mesh fencing with detachable MS Square section poles of 1.7 Meter Height & heavy foot plate for fastening, doors on both Drive and Non-Drive side, including Safety switch interlock, Lamp indications, Boom Barrier, Hooters and Light curtains interlocks. • Electromagnetic actuators with interlocks for safety access control at doors. • All fasteners should be "captive type" to avoid loss of fasteners during dismantling. • 2" MS Pipe detachable barricading of 1.5 Meter Height with Hinge door at Uncoiler, Turnstile, Pit area equipment as frequent coil/packet handling
Pune SC	CRNCTL Line, CRS Line, CRWCTL Line, HRNCTL Line, HRS Line, HRWCTL Line, CR Mini CTL Line, CR Trapezoidal Line, Transfer trolley, Solar Inverter room fencing	10 Set	
Indore SC	HRS Line, HRWCTL Line, Transfer trolley	3 Set	
Chennai SC	CRNCTL Line, CRS Line, CRWCTL Line, HRNCTL Line, HRWCTL Line, HRS Line, Transfer trolley, Solar Inverter room fencing	9 Set	

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			will be done through crane. • 3" MS Pipe detachable barricading of 1 Meter Height for coil transfer trolley with 2 Nos 1000 mm Boom Barrier for entrance as frequent coil handling will be done through crane. • Safety Fencing of 1.5 Mtr Height with 2 Nos hinge doors for Solar inverter room

BOQ FORMAT FOR DESIGN, ENGINEERING, SUPPLY, ERECTION, INSTALLATION, COMMISSIOING OF STEEL MESH SAFETY FENCING FOR PROCESS LINES OF BAHADURGARH SC

SR. NO.	LINE	PRODUCT	QTY	UNIT
Cold Rolled Slitter Line				
1	CRS	Steel Mesh Safety Fence	76	Mtr
2	CRS	2" MS Pipe barricaded structure	32	Mtr
3	CRS	Safety hinge door	6	Set
4	CRS	Safety Magnetic switch	6	Set
5	CRS	Boom barrier 3000 mm for Uncoiler & Recoiler coil car	2	Set
6	CRS	Boom barrier 1000 mm for Slitter arbour area	1	Set
7	CRS	Safety control panel with power supply & safety relay/PLC	1	Set
8	CRS	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
Hot Rolled Slitter Line				
1	HRS	Safety Fence	90	Mtr
2	HRS	2" MS Pipe barricaded structure	42	Mtr
3	HRS	Safety hinge door	6	Set
4	HRS	Safety Magnetic switch	6	Set
5	HRS	Boom barrier 3500 mm for Uncoiler & Recoiler coil car	2	Set
6	HRS	Boom barrier 1800 mm for Slitter arbour area	1	Set
7	HRS	Safety control panel with power supply & safety relay/PLC	1	Set

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8	HRS	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
CR Wider Cut to length Line				
1	CRWCTL	Safety Fence	100	Mtr
2	CRWCTL	2" MS Pipe barricaded structure	17	Mtr
3	CRWCTL	Safety hinge door	5	Set
4	CRWCTL	Safety Magnetic switch	5	Set
5	CRWCTL	Boom barrier 3000 mm for Uncoiler coil car	1	Set
6	CRWCTL	Boom barrier 3000 mm for Piler-1,2	4	Set
7	CRWCTL	Safety control panel with power supply & safety relay/PLC	1	Set
8	CRWCTL	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
CR Narrow Cut to length Line				
1	CRNCTL	Safety Fence	83	Mtr
2	CRNCTL	2" MS Pipe barricaded structure	15	Mtr
3	CRNCTL	Safety hinge door	5	Set
4	CRNCTL	Safety Magnetic switch	5	Set
5	CRNCTL	Boom barrier 3000 mm for Uncoiler coil car	1	Set
6	CRNCTL	Boom barrier 2500 mm for Piler-1,2	4	Set
7	CRNCTL	Safety control panel with power supply & safety relay/PLC	1	Set
8	CRNCTL	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
HR Narrow Cut to length Line				
1	HRNCTL	Safety Fence	90	Mtr
2	HRNCTL	2" MS Pipe barricaded structure	17	Mtr
3	HRNCTL	Safety hinge door	5	Set
4	HRNCTL	Safety Magnetic switch	5	Set
5	HRNCTL	Boom barrier 3000 mm for Uncoiler coil car	1	Set
6	HRNCTL	Boom barrier 2500 mm for Piler-1,2	4	Set
7	HRNCTL	Safety control panel with power supply & safety relay/PLC	1	Set
8	HRNCTL	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
Transfer trolley-1 & 2				
1	Transfer Trolley - 1	3" Heavy duty MS round pipe barricaded Structure with 1 meter in Height	128	Mtr
2		Boom barrier 1000 mm for allowing crossing	2	Nos
3		Safety control panel with power supply & safety relay/PLC	1	Set
4		Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
1	Transfer Trolley - 2	3" Heavy duty MS round pipe barricaded Structure with 1 meter in Height	60	Mtr
2		Boom barrier 1000 mm for allowing crossing	2	Nos
3		Safety control panel with power supply & safety relay/PLC	1	Set
4		Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
Solar Inverter Panel Area				
1	Solar Inverter	Safety fencing with 1.5 Meter height with 2 Nos Hinge doors with lock & key mechanism	40	Mtr

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BOQ FORMAT FOR DESIGN, ENGINEERING, SUPPLY, ERECTION, INSTALLATION, COMMISSIOING OF STEEL MESH SAFETY FENCING FOR PROCESS LINES OF PUNE SC

SR. NO.	LINE	PRODUCT	QTY	UNIT
		Cold Rolled Slitter Line		
1	CRS	Steel Mesh Safety Fence	50	Mtr
2	CRS	2" MS Pipe barricaded structure	25	Mtr
3	CRS	Safety hinge door	1	Set
4	CRS	Safety Sliding door	2	Set
5	CRS	Safety Magnetic switch	3	Set
6	CRS	Boom barrier 4000 mm for Uncoiler & Recoiler coil car	2	Set
7	CRS	Boom barrier 1000 mm for Slitter arbour area	1	Set
8	CRS	Boom barrier 1000 mm for Scrap Trolley	1	
9	CRS	Safety control panel with power supply & safety relay/PLC	1	Set
10	CRS	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
		Hot Rolled Slitter Line		
1	HRS	Safety Fence	90	Mtr
2	HRS	2" MS Pipe barricaded structure	20	Mtr
3	HRS	Safety hinge door	2	Set
4	HRS	Safety Sliding door	1	Set
5	HRS	Safety Magnetic switch	3	Set
6	HRS	Boom barrier 4000 mm for Uncoiler & Recoiler coil car	2	Set
7	HRS	Boom barrier 1500 mm for Slitter arbour area	1	Set
8	HRS	Safety control panel with power supply & safety relay/PLC	1	Set
8	HRS	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
		CR Wider Cut to length Line		
1	CRWCTL	Safety Fence	105	Mtr
2	CRWCTL	2" MS Pipe barricaded structure	30	Mtr
3	CRWCTL	Safety hinge door	1	Set
4	CRWCTL	Safety Sliding door	2	Set
5	CRWCTL	Safety Magnetic switch	5	Set
6	CRWCTL	Boom barrier 4000 mm for Uncoiler coil car	1	Set
7	CRWCTL	Boom barrier 3000 mm for Piler-1,2	4	Set
8	CRWCTL	Safety control panel with power supply & safety relay/PLC	1	Set
9	CRWCTL	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
		CR Trapezoidal Cut to length Line		
1	CRTRAPCTL	Safety Fence	95.9	Mtr
2	CRTRAPCTL	2" MS Pipe barricaded structure	46.1	Mtr
3	CRTRAPCTL	Safety hinge door	2	Set
4	CRTRAPCTL	Safety Sliding door	1	Set

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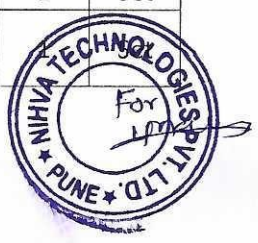
5	CRTRAPCTL	Safety Magnetic switch	3	Set
6	CRTRAPCTL	Boom barrier 3000 mm for Uncoiler coil car	1	Set
7	CRTRAPCTL	Boom barrier 3000 mm for Piler-1,2	4	Set
8	CRTRAPCTL	Safety control panel with power supply & safety relay/PLC	1	Set
9	CRTRAPCTL	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
CR Narrow Cut to length Line				
1	CRNCTL	Safety Fence	50	Mtr
2	CRNCTL	2" MS Pipe barricaded structure	21	Mtr
3	CRNCTL	Safety hinge door	1	Set
4	CRNCTL	Safety Slinding door	1	Set
5	CRNCTL	Safety Magnetic switch	2	Set
6	CRNCTL	Boom barrier 3000 mm for Uncoiler coil car	1	Set
7	CRNCTL	Boom barrier 4000 mm for Piler-1,2	2	Set
8	CRNCTL	Safety control panel with power supply & safety relay/PLC	1	Set
9	CRNCTL	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
CR Mini Cut to length Line				
1	CR MINI CTL	Safety Fence	30	Mtr
3	CR MINI CTL	Safety hinge door	2	Set
4	CR MINI CTL	Safety Magnetic switch	2	Set
5	CR MINI CTL	Boom barrier 2000 mm for Uncoiler coil car	1	Set
8	CR MINI CTL	Safety control panel with power supply & safety relay/PLC	1	Set
9	CR MINI CTL	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
HR Narrow Cut to length Line				
1	HRNCTL	Safety Fence	80	Mtr
2	HRNCTL	2" MS Pipe barricaded structure	24	Mtr
3	HRNCTL	Safety hinge door	4	Set
4	HRNCTL	Safety Magnetic switch	4	Set
5	HRNCTL	Boom barrier 4000 mm for Uncoiler coil car	1	Set
6	HRNCTL	Boom barrier 2500 mm for Piler-1,2	4	Set
7	HRNCTL	Boom barrier 4000 mm for Scrap Car Trolly	1	Set
8	HRNCTL	Safety control panel with power supply & safety relay/PLC	1	Set
9	HRNCTL	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
HR Wider Cut to length Line				
1	HRWCTL	Safety Fence	78	Mtr
2	HRWCTL	2" MS Pipe barricaded structure	43	Mtr
3	HRWCTL	Safety hinge door	3	Set
4	HRWCTL	Safety Magnetic switch	3	Set
5	HRWCTL	Boom barrier 4000 mm for Uncoiler coil car	1	Set
6	HRWCTL	Boom barrier 3000 mm for Leveller	1	Set

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7	HRWCTL	Boom barrier 2000 mm for Leveller	1	Set
9	HRWCTL	Safety control panel with power supply & safety relay/PLC	1	Set
10	HRWCTL	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
		Coil Transfer trolley		
	Coil Transfer Trolley	3" Heavy duty MS round pipe barricaded Structure with 1 meter in Height	74	Mtr
	Coil Transfer Trolley	Boom barrier 2000 mm for allowing crossing	2	Set
	Coil Transfer Trolley	Safety control panel with power supply & safety relay/PLC	1	Set
	Coil Transfer Trolley	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	Set
1	Solar Inverter room	Safety fencing with 1.5 Meter height with 2 Nos Hinge doors with lock & key mechanism	40	Mtr

BOQ FORMAT FOR DESIGN, ENGINEERING, SUPPLY, ERECTION, INSTALLATION, COMMISSIOING OF STEEL MESH SAFETY FENCING FOR PROCESS LINES OF CHENNAI SC

SR. NO.	LINE	PRODUCT	QTY	UNIT
		Cold Rolled Slitter Line		
1	CRS	Steel Mesh Safety Fence	76	Mtr
2	CRS	2" MS Pipe barricaded structure	32	Mtr
3	CRS	Safety hinge door	6	Set
4	CRS	Safety Magnetic switch	6	Set
5	CRS	Boom barrier 3000 mm for Uncoiler & Recoiler coil car	2	Set
6	CRS	Boom barrier 1000 mm for Slitter arbour area	1	Set
7	CRS	Safety control panel with power supply & safety relay/PLC	1	Set
8	CRS	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
		Hot Rolled Slitter Line		
1	HRS	Safety Fence	90	Mtr
2	HRS	2" MS Pipe barricaded structure	42	Mtr
3	HRS	Safety hinge door	6	Set
4	HRS	Safety Magnetic switch	6	Set
5	HRS	Boom barrier 3500 mm for Uncoiler & Recoiler coil car	2	Set
6	HRS	Boom barrier 1800 mm for Slitter arbour area	1	Set
7	HRS	Safety control panel with power supply & safety relay/PLC		



8	HRS	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
CR Wider Cut to Length Line				
1	CRWCTL	Safety Fence	100	Mtr
2	CRWCTL	2" MS Pipe barricaded structure	17	Mtr
3	CRWCTL	Safety hinge door	5	Set
4	CRWCTL	Safety Magnetic switch	5	Set
5	CRWCTL	Boom barrier 3000 mm for Uncoiler coil car	1	Set
6	CRWCTL	Boom barrier 3000 mm for Piler-1,2	4	Set
7	CRWCTL	Safety control panel with power supply & safety relay/PLC	1	Set
8	CRWCTL	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
CR Narrow Cut to Length Line				
1	CRNCTL	Safety Fence	83	Mtr
2	CRNCTL	2" MS Pipe barricaded structure	15	Mtr
3	CRNCTL	Safety hinge door	5	Set
4	CRNCTL	Safety Magnetic switch	5	Set
5	CRNCTL	Boom barrier 3000 mm for Uncoiler coil car	1	Set
6	CRNCTL	Boom barrier 2500 mm for Piler-1,2	4	Set
7	CRNCTL	Safety control panel with power supply & safety relay/PLC	1	Set
8	CRNCTL	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
HR Narrow Cut to Length Line				
1	HRNCTL	Safety Fence	90	Mtr
2	HRNCTL	2" MS Pipe barricaded structure	17	Mtr
3	HRNCTL	Safety hinge door	5	Set
4	HRNCTL	Safety Magnetic switch	5	Set
5	HRNCTL	Boom barrier 3000 mm for Uncoiler coil car	1	Set
6	HRNCTL	Boom barrier 2500 mm for Piler-1,2	4	Set
7	HRNCTL	Safety control panel with power supply & safety relay/PLC	1	Set
8	HRNCTL	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
1	HRWCTL	Safety Fence	80	Mtr
2	HRWCTL	2" MS Pipe barricaded structure	18	Mtr
3	HRWCTL	Sliding Doors of 1000 mm Width	5	Nos
4	HRWCTL	Hinged Doors of 800 mm width	2	Nos
5	HRWCTL	Boom Barrier of 3500 mm width	1	No
6	HRWCTL	Light Curtain (Leveller movement, Width 2000mm	1	No
Coil Transfer trolley				
1	Coil Transfer Trolley	3" Heavy duty MS round pipe barricaded Structure with 1 meter in Height	230	Mtr

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2	Coil Transfer Trolley	Boom barrier 1000 mm for allowing crossing	2	Set
3	Coil Transfer Trolley	Safety control panel with power supply & safety relay/PLC	1	Set
4	Coil Transfer Trolley	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	Set
Solar Inverter Panel Area				
1	Solar Inverter Panels	Safety fencing with 1.5 Meter height with 2 Nos Hinge doors with lock & key mechanism	40	Mtr

BOQ FORMAT FOR DESIGN, ENGINEERING, SUPPLY, ERECTION, INSTALLATION, COMMISSIOING OF STEEL MESH SAFETY FENCING FOR PROCESS LINES OF INDORE SC

SR. NO.	LINE	PRODUCT	QTY	UNIT
Hot Rolled Slitter Line				
1	HRS	Safety Fence	90	Mtr
2	HRS	2" MS Pipe barricaded structure	15	Mtr
3	HRS	Safety hinge door	7	Set
4	HRS	Safety Magnetic switch	7	Set
5	HRS	Boom barrier 3500 mm for Uncoiler & Recoiler coil car	2	Set
6	HRS	Boom barrier 1800 mm for Slitter arbour area	1	Set
7	HRS	Safety control panel with power supply & safety relay/PLC	1	Set
8	HRS	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
HR Wider Cut to length line				
1	HRWCTL	Safety Fence	105	Mtr
3	HRWCTL	Safety hinge door	7	Set
4	HRWCTL	Safety Magnetic switch	7	Set
5	HRWCTL	Boom barrier 3500 mm for Uncoiler coil car	1	Set
6	HRWCTL	Boom barrier 1800 mm for Leveller movement	1	Set
7	HRWCTL	Light Curtain 8000 mm range with protection guard	2	Set
8	HRWCTL	Safety control panel with power supply & safety relay/PLC	1	Set
9	HRWCTL	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	LOT
Coil Transfer Trolley				
1	Coil Transfer Trolley	3" Heavy duty MS round pipe barricaded Structure with 1 meter in Height	40	Mtr
2	Coil Transfer Trolley	Boom barrier 1000 mm for allowing crossing	2	Set

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3	Coil Transfer Trolley	Safety control panel with power supply & safety relay/PLC	1	Set
4	Coil Transfer Trolley	Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing	1	Set

AMNS SAFETY RULE AND REGULATION AS PER COMPANY POLICY

1) All necessary safety PPE's i.e. safety shoes, leather gloves, goggles, masks etc. in contractor scope
2) All work to be carried out as per safety norms which shall be guided to AM/NS team during work & no tolerance in safety will be accepted and penalty to be imposed if any violation.
3) Contractor shall not proceed with next activity until previous activity will be checked and approved by Representative.
4) All required tools & tackles will provide by contractor
5) Contractor shall deploy and provide one full time engineer to supervise the project and has experienced of carrying out such type of work.
6) Contractor will provide all required consumable items for the job
7) Contractor shall be responsible for procuring, supplying, transporting, and providing all labor, materials, tools and equipment etc., required for completion of the work.
8) All materials shall be stored in a proper manner protected from natural elements so as to avoid contamination and deterioration.
9) Workers working on the site shall be skilled in their job and have related job experience.
10) The contractor shall guarantee that all work performed will be free from all defects in workmanship and materials and that all installation will provide the capacities and characteristics specified. The contract further guarantees that if, during a period of one year from the date of the certificate of completion and acceptance of the work, any such defects will be repaired by the contractor at his own cost.

AMNS Scope:

1) Electricity supply
2) On site help i.e. Workshop Facilities etc. in AM/NS scope

